PROJECT REPORT

📄 Salary Predictor using Linear Regression

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Project Title: Salary Predictor using Linear Regression

Course: Fundamental of AIML (CSA2001)

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2. Introduction

Salary prediction is a common real-world problem in data analysis and machine learning. In many organizations, salary often increases with experience, making it possible to estimate salary using historical data.

This project uses:

- **Python** for implementation
- **Pandas** for data handling
- **NumPy** for numerical operations
- **Matplotlib** for visualization
- **Scikit-learn** for building the Linear Regression model

The dataset used is Salary_Data.csv, which contains:

- **YearsExperience** → independent variable
- **Salary** → dependent variable

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3. Objective

The objective of this project is to predict an employee's salary based on their years of experience using a **Linear Regression** model. The project demonstrates how machine learning can be used to identify and model a linear relationship between input and output variables.

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4. Problem Statement

To build a machine learning model that predicts salary based on the number of years of experience using a simple linear regression algorithm.

5.Key Decisions Made

- Chose **salary prediction** as the project problem because it is simple and practical.
- Selected **Simple Linear Regression** since the data has one input (YearsExperience) and one output (Salary).
- Used **Python** for implementation because it is easy and suitable for machine learning projects.
- Used **Pandas** to load the dataset and **Scikit-learn** to train the model.
- Defined YearsExperience as the **input** and Salary as the **output**.
- Reshaped the input data to meet Scikit-learn requirements.
- Added **user input** to make the project interactive.
- Used **try-except** to handle invalid input.
- Included **graph plotting** to visualize the data and regression line.
- Kept the project simple without advanced evaluation methods.

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6. Dataset Description

The dataset file Salary_Data.csv contains two columns:

Column Name	Description
YearsExperience	Number of years of work experience
Salary	Salary corresponding to the experience

This is a **simple supervised learning dataset** where:

- Input feature = YearsExperience
- Target output = Salary

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7. Methodology

The project follows these steps:

1. **Load the dataset**
    - Read the CSV file using Pandas.

2. **Prepare input and output**
    - Extract YearsExperience as input variable a
    - Extract Salary as output variable b

3. **Train the model**
    - Create a LinearRegression() object
    - Fit the model on the dataset

4. **Extract model parameters**
    - Slope (m) using model.coef_
    - Intercept (c) using model.intercept_

5. **Prediction**
    - Accept years of experience from the user
    - Predict salary using the trained model

6. **Visualization**
    - Plot actual data points using scatter plot
    - Plot regression line using the model predictions

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8. Algorithm Used

Linear Regression

Linear Regression is a supervised machine learning algorithm used to model the relationship between a dependent variable and one or more independent variables.

The equation of simple linear regression is:

$$Y = mX + c$$

Where:

- Y = Predicted salary
- X = Years of experience
- m = Slope of the line
- c = Intercept

In this project:

- Years Experience is the predictor
- Salary is the target variable

9. Challenges Faced

While implementing the project, some challenges were faced.

8.1 Reshaping the input data

At first, it was necessary to understand why the experience values had to be reshaped. Scikit-learn requires input features in a 2D format, so this was solved using:

reshape(-1,1)

8.2 Handling invalid input

Users may enter text or invalid values instead of numbers. To avoid program errors, a try-except block was used.

```
try:
  x = float(input("Enter experience: "))
except:
  print("Invalid input")
```

8.3 Limited dataset

The dataset only contains one feature, so the prediction is based solely on experience. This makes the model simple but less realistic than real-world salary prediction systems.

10.Code Screenshot

```python
# Salary Prediction using Linear Regression

import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.linear_model import LinearRegression

#Load dataset
data = pd.read_csv("Salary_Data.csv")
# simple dataset
a=data["YearsExperience"].values.reshape(-1,1)
b=data["Salary"].values

# create model
model = LinearRegression()
model.fit(a, b)

# checking equation
m = model.coef_[0]
c = model.intercept_

print("Model trained")
print("Equation: Salary =", m, "* Experience +", c)

# take user input
try:
    x = float(input("Enter experience: "))
    pred = model.predict([[x]])
    print("Predicted Salary: ₹", int(pred[0]))
except:
    print("Invalid input")

# plotting
plt.scatter(a,b)
plt.plot(a, model.predict(a))

plt.xlabel("Experience")
plt.ylabel("Salary")
plt.title("Salary Prediction")

plt.show()
```

11.Output Screenshot

```
Model trained
Equation: Salary = 4793.939393939394 * Experience + 25333.333333333332
Enter experience:  7
Predicted Salary: ₹ 58890
```

12. <u>What I Learned</u>

This project helped me learn both technical and practical concepts of machine learning.

Technical learning:

- How to read a CSV file using Pandas
- How to separate input and output variables
- How to use Scikit-learn to train a linear regression model
- How to make predictions from a trained model
- How to visualize data using Matplotlib

Conceptual learning:

- Understanding regression problems
- Understanding the formula of linear regression
- Knowing how slope and intercept affect prediction
- Learning how machine learning can model relationships in data

Practical learning:

- The importance of preparing data correctly
- The value of handling user errors
- The usefulness of visualization in understanding results
- The need for evaluation metrics in machine learning projects

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13. <u>Result</u>

The Linear Regression model successfully predicts salary based on years of experience. The plotted graph shows a positive linear relationship between experience and salary, indicating that salary tends to increase with work experience.

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14. Future Enhancements

The project can be improved by:

- Splitting the dataset into training and testing sets
- Evaluating accuracy using regression metrics
- Adding more features such as:
    - Education
    - Job role
    - Location
    - Skills
- Saving the trained model for later use
- Building a GUI or web application for user interaction

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15. Conclusion

This project demonstrates the implementation of a **Simple Linear Regression** model to predict salary based on years of experience. It is a good beginner-level machine learning project that helps understand model training, prediction, and visualization. The project clearly shows how data can be used to make meaningful predictions in practical scenarios.